CONJECTURE 4.2 OF ALTINIŞIK AND ALTINTAŞ DOES NOT EXTEND TO TEN ELEMENTS

Abstract. Altınışık and Altıntaş conjectured that a singular LCM matrix on a nine-element gcd-closed set forces a cube sublattice. Mattila, Haukkanen, and Mäntysalo proved the nine-element statement and left open whether its conclusion extends to cardinalities $10, 11, 12$. We give a ten-element gcd-closed set whose LCM matrix is singular although its divisibility poset contains no copy of the Boolean lattice B_3. An elementary stabilization yields such sets in every cardinality $n \geq 10$, and known results for $n \leq 9$ show that ten is sharp.

1. Introduction

For a finite set $S = \{x_1, \ldots, x_n\}$ of distinct positive integers, write

$$[S] = \left(\operatorname{lcm}(x_i, x_j)\right)_{i,j=1}^{n}.$$

The set S is *gcd-closed* if $\gcd(x_i, x_j) \in S$ for all i, j. We call $(S, |)$ B_3*-free* if no subset of S, with its inherited divisibility order, is isomorphic to the eight-element Boolean lattice B_3. In particular, a B_3-free poset contains no cube subsemilattice.

Conjecture 4.2 of Altınışık and Altıntaş [1] states that if $|S| = 9$ and $[S]$ is singular, then $(S, |)$ contains a cube sublattice. Mattila, Haukkanen, and Mäntysalo proved the nine-element statement, produced a thirteen-element counterexample to its cardinality extension, and explicitly left the cases $10, 11, 12$ open [4]. The result below settles those cases. It is a counterexample to the cardinality extension of Conjecture 4.2, not to its literal nine-element formulation.

Theorem 1. *For an integer $n \geq 1$, there exists an n-element gcd-closed set S such that $[S]$ is singular and $(S, |)$ is B_3-free if and only if $n \geq 10$.*

2. The counterexample

Let

$$S_0 = \{1, 2, 3, 5, 7, 14, 15, 35, 102, 3570\}, \qquad T = 3570.$$

Set

$$A = \{2, 3, 5, 7\}, \qquad M = \{14, 15, 35, 102\}.$$

The elements of M lie above the atom pairs

$$\{2, 7\}, \quad \{3, 5\}, \quad \{5, 7\}, \quad \{2, 3\},$$

2020 *Mathematics Subject Classification.* Primary 11C20; Secondary 15B36, 06A12.
Key words and phrases. LCM matrix, gcd-closed set, Boolean lattice, meet semilattice.

respectively. Their atom-pair graph is the cycle $2 - 3 - 5 - 7 - 2$.

Proposition 2. *The set S_0 is gcd-closed, the matrix $[S_0]$ is singular, and $(S_0, |)$ is B_3-free.*

Proof. Since

$$102 = 2 \cdot 3 \cdot 17, \qquad T = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 17,$$

every element of S_0 divides T. Distinct elements of A have gcd 1, and for $p \in A$ and $m \in M$ one has $\gcd(p, m) = p$ if $p \mid m$ and 1 otherwise. Among the elements of M,

$$\gcd(14, 15) = 1, \quad \gcd(14, 35) = 7, \quad \gcd(14, 102) = 2,$$
$$\gcd(15, 35) = 5, \quad \gcd(15, 102) = 3, \quad \gcd(35, 102) = 1.$$

Thus S_0 is gcd-closed.

In the displayed order of S_0, let

$$v = (-3570, 1785, 1190, 714, 510, -255, -238, -102, -35, 1)^{\mathsf{T}}.$$

Equivalently,

$$v_x = \varepsilon_x \frac{T}{x}, \qquad \varepsilon_x = \begin{cases} -1, & x \in \{1\} \cup M, \\ 1, & x \in A \cup \{T\}. \end{cases}$$

For $s \in S_0$,

$$([S_0]v)_s = sT \sum_{x \in S_0} \frac{\varepsilon_x}{\gcd(s, x)}.$$

The sum on the right is zero for every row type. Indeed,

$$-1 + 4 - 4 + 1 = 0$$

when $s = 1$. If $s = p \in A$, then, since every atom has degree two in the atom-pair graph, the sum is

$$-1 + \left(3 + \frac{1}{p}\right) - \left(2 + \frac{2}{p}\right) + \frac{1}{p} = 0.$$

If $s = m \in M$ lies above $p, q \in A$, it is

$$-1 + \left(2 + \frac{1}{p} + \frac{1}{q}\right) - \left(1 + \frac{1}{p} + \frac{1}{q} + \frac{1}{m}\right) + \frac{1}{m} = 0.$$

Finally, for $s = T$ it is

$$-1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} - \frac{1}{14} - \frac{1}{15} - \frac{1}{35} - \frac{1}{102} + \frac{1}{3570} = 0,$$

as is seen after multiplication by 3570:

$$-3570 + 4199 - 630 + 1 = 0.$$

Hence $[S_0]v = 0$, so $[S_0]$ is singular.

The divisibility poset has four levels,

$$\{1\}, \qquad A, \qquad M, \qquad \{T\}.$$

Every element of B_3 lies on a maximal chain of four elements, and every four-element chain in $(S_0, |)$ uses one element from each displayed level. Therefore any copy of B_3 in $(S_0, |)$ would use 1 and T as its bottom and top, three elements of A as its atoms, and three elements of M as its coatoms. Those coatoms would represent the three pairs of the selected atoms and hence would form a triangle in the atom-pair graph. That graph is the triangle-free cycle C_4. Thus $(S_0, |)$ is B_3-free. □

3. Larger orders and sharpness

The singularity-preserving construction below goes back to Hong [3]; we record that it also preserves B_3-freeness.

Lemma 3. *Let S be a finite gcd-closed set containing 1, and let $a > 1$ be an integer. Put*

$$\sigma_a(S) = \{1\} \cup aS.$$

If $[S]$ is singular, then $[\sigma_a(S)]$ is singular. If $(S, |)$ is B_3-free, then so is $(\sigma_a(S), |)$.

Proof. Gcd-closure follows from $\gcd(1, ax) = 1$ and $\gcd(ax, ay) = a\gcd(x, y)$, and $|\sigma_a(S)| = |S| + 1$. Write $S = \{x_1, \ldots, x_n\}$ with $x_1 = 1$ and $x = (x_1, \ldots, x_n)^{\mathsf{T}}$. In the order $1, ax_1, \ldots, ax_n$,

$$[\sigma_a(S)] = \begin{pmatrix} 1 & ax^{\mathsf{T}} \\ ax & a[S] \end{pmatrix}.$$

Choose $0 \neq z \in \ker[S]$. The row of $[S]$ indexed by 1 is x^{T}, so $x^{\mathsf{T}}z = 0$. Consequently,

$$[\sigma_a(S)] \begin{pmatrix} 0 \\ z \end{pmatrix} = 0.$$

Suppose that $Q \subseteq \sigma_a(S)$ is a copy of B_3. If $1 \notin Q$, then $Q \subseteq aS$, and division by a gives a copy of B_3 in S. If $1 \in Q$, then 1 is the bottom of Q. The element a cannot belong to Q: within Q it would have six strict upper bounds, whereas a nonbottom element of B_3 has at most three. Thus

$$Q' = (Q \setminus \{1\}) \cup \{a\}$$

is a copy of B_3 in aS, and division by a gives one in S. Thus B_3-freeness is preserved. □

Proof of Theorem 1. Proposition 2 gives the case $n = 10$, and iterating Lemma 3 gives every $n \geq 10$.

For the converse, Hong proved that every LCM matrix on a gcd-closed set with at most seven elements is nonsingular [3]. Haukkanen, Mattila, and Mäntysalo proved that an eight-element singular example has divisibility poset isomorphic to B_3 [2]. Mattila, Haukkanen, and Mäntysalo proved that every nine-element singular example contains a cube subsemilattice [4]. Consequently no B_3-free singular example has fewer than ten elements. □

References

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